English

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ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2024-2025)

Test Pattern

NEET (UG)

MAJOR

27-04-2025

PRE-MEDICAL : ENTHUSIAST, LEADER & ACHIEVER COURSE PHASE - ALL PHASE

Test Booklet Code

This Booklet contains **28** pages.

L17

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : ____________________

Form Number : in figures ____________________

: in words ____________________

Centre of Examination (in Capitals) : ____________________

Candidate's Signature : ____________________ **Invigilator's Signature :** ____________________

YOUR TARGET IS TO SECURE GOOD RANK IN PRE-MEDICAL 2025

SUBJECT : BIOLOGY

Topic : FULL SYLLABUS.

1. Which among the following is not a direct cause of biodiversity loss?
A. Introduction of invasive species
B. Over exploitation
C. Mutualism
D. Habitat loss
Choose the correct option :
(1) A, B and D (2) A, B and C
(3) C only (4) D only

2. Match the taxonomic group of animal in Column-I with their percentage of threat of extinction in Column-II and choose the correct option from the codes given below :

	Column-I		Column-II
A.	Birds	i.	12%
B.	Amphibians	ii.	23%
C.	Mammals	iii.	31%
D.	Gymnosperm	iv.	32%

(1) A-i, B-ii, C-iii, D-iv
(2) A-ii, B-i, C-iv, D-iii
(3) A-i, B-iv, C-ii, D-iii
(4) A-iv, B-iii, C-ii, D-i

3. Given below are two statements :
Statement I : According to Gause's Competitive Exclusion Principle, two species occupying the same niche can coexist indefinitely in a stable environment.
Statement II : If two species compete for exactly the same resources, one can be excluded due to competitive disadvantage.
Choose the correct answer from the options given below :
(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

4. Given below are two statements.
Statement-I : PCT is lined by simple cuboidal brush border epithelium which increases the surface area for reabsorption.
Statement-II : Conditional reabsorption of Na^+ & water takes place in DCT.
(1) Both Statement-I and Statement-II are true.
(2) Both Statement-I and Statement-II are false.
(3) Statement-I is true but statement II is false.
(4) Statement I is false but statement II is true.

5. Choose the correct statement given below regarding nephron.
(1) The DCTs of many nephrons open into a straight tube called collecting duct.
(2) The malpighian corpuscle, PCT and DCT of the nephron are situated in the cortical region of the kidney.

(3) The efferent arteriole emerging from the glomerulus forms a fine capillary network around the renal tubule called the peritubular capillaries.
(4) All the above

6. Match List-I with List-II

	List-I		List-II
(A)	Endomembrane system	(I)	Nucleolus
(B)	r-RNA synthesis	(II)	Golgi body
(C)	Circular DNA	(III)	Cilia and flagella
(D)	9 + 2 Arrangement	(IV)	Chloroplast

Choose the correct answer from the options given below :
(1) (A–IV), (B–III), (C–II), (D–I)
(2) (A–IV), (B–II), (C–III), (D–I)
(3) (A–II), (B–IV), (C–I), (D–III)
(4) (A–II), (B–I), (C–IV), (D–III)

7. Match List-I with List-II

	List-I		List-II
(A)	Zygotene	(I)	Crossing over
(B)	Diplotene	(II)	Synapsis
(C)	Pachytene	(III)	Terminalisation of chiasmata
(D)	Diakinesis	(IV)	Chiasmata formation

Choose the correct answer from the options given below :

(1) (A–IV), (B–II), (C–III), (D–I)

(2) (A–I), (B–II), (C–IV), (D–III)

(3) (A–II), (B–IV), (C–I), (D–III)

(4) (A–II), (B–I), (C–III), (D–IV)

8. Following are the stages of cell division?
(A) Synthetic phase
(B) Anaphase
(C) Cytokinesis
(D) Metaphase
(E) G_2 – Phase
Choose the correct sequence of stages from the options given below-

(1) C–E–D–A–B (2) A–E–D–B–C

(3) B–C–A–D–E (4) D–E–C–B–A

9. Given below are two statements :-
Statement-I : Mitochondria and chloroplast both have 70s ribosome.
Statement-II : The ribosomes of the chloroplasts are larger than the cytoplasmic ribosome.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement-I and Statement-II are correct

(2) Both Statement-I and Statement-II are incorrect

(3) Statement-I is correct but Statement-II is incorrect

(4) Statement-I is incorrect but Statement-II is correct

10. Which cell organelles have double stranded circular DNA molecule and ribosome-

(1) Chloroplast (2) Centriole

(3) Golgi body (4) Lysosomes

11. Given below are two statements-
Statement-I : Compaction of chromosomes continues throughout leptotene.
Statement-II : 'X' shaped structure are called chiasmata which is visible in diplotene.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement-I and Statement-II are true

(2) Both Statement-I and Statement-II are false

(3) Statement-I is true but Statement-II is false

(4) Statement-I is false but Statement-II is true

12. At which stage all chromosomes coming to lie at the equator with one chromats of each chromosomes connected by its kinetochore to spindle fibre from one pole and its sister chromatid connected by its kinetochore to spindle fibre from the opposite pole -

(1) Prophase (2) Metaphase

(3) Anaphase (4) Telophase

13. The fate of a piece of DNA, carrying only gene of interest which is transferred into an alien organism :-
(A) This piece of DNA, would not be able to multiply itself in the progency cell of organism.

(B) It may multiply and be inherited along with the host DNA.
(C) The alien piece of DNA never become part of chromosome.
(D) It shows ability to self replicate.
Choose the correct answer regarding above statement from the options given below :-

(1) A and B only (2) C and D only

(3) A & C only (4) B & C only

14. Given below are two statements :-
Statement I :- The choice of Bt-genes depends upon the targeted pest only.
Statement II :- The protein encoded by the genes cry I Ab controls corn borer.
In the light of the above statements, choose the correct answer from the options given below :-

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true

15. Match List-I with List-II

List-I		List-II	
A.	*Monascus purpureus*	I.	Citric acid
B.	*Aspergillus niger*	II.	Cyclosporin A
C.	*Acetobacter aceti*	III.	Statins
D.	*Trichoderma polysporum*	IV.	Acetic acid

Choose the correct answer from the options given below :

(1) A-III, B-I, C-II, D-IV

(2) A-II, B-IV, C-III, D-I

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-I, C-III, D-II

16. The recognition sequence of EcoRI consists of :-

(1) 8 bp (2) 6 bp

(3) 4 bp (4) 10 bp

17. Method of producing thousands of plants through tissue culture in very short duration is called :-

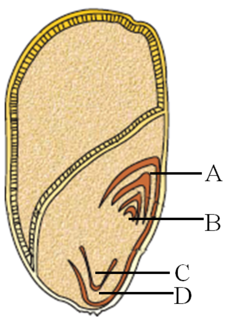
(1) Totipotency

(2) Micropropagation

(3) Re-differentiation

(4) Pluripotency

18. Identify the part of the seed from the given figure which is destind to form shoot when the seed germinates.



(1) A (2) B (3) C (4) D

19. Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figure (a) and (b) :-

(a) (b)

(1) (a) Epigynous, (b) Hypogynous

(2) (a) Hypogynous, (b) Epigynous

(3) (a) Hypogynous, (b) Perigynous

(4) (a) Perigynous, (b) Perigynous

20. Which of the following is not an example of actinomorphic flower ?

(1) Datura (2) *Sesbania*

(3) Mustard (4) Brinjal

21. Match List-I with List-II :-

List-I		List-II	
(A)	Cucumber	(I)	Imbricate aestivation
(B)	Lemon	(II)	Epigynous flower
(C)	Cassia	(III)	Drupe
(D)	Coconut	(IV)	Axile placentation

Choose the correct answer from the options given below.

(1) A-II, B-IV, C-I, D-III

(2) A-I, B-II, C-III, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-III, C-IV, D-I

22. Somatic hybrid plants are produced by the fusion of

(1) Embryo

(2) Pollengrain

(3) Roots

(4) Protoplast

23. Match List-I with List-II

	List-I		List-II
(A)	Human alpha-lactalbumin	(i)	HIV
(B)	P.C.R.	(ii)	Cotton bollworms
(C)	Cry-II Ab	(iii)	ADA-deficiency
(D)	Bone marrow transplantation	(iv)	Rosie cow milk

Choose the correct answer from the option given below :

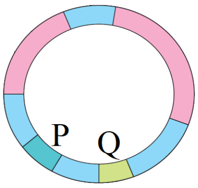
(1) A-(iv), B-(i), C-(iii), D-(ii)

(2) A-(iv), B-(i), C-(ii), D-(iii)

(3) A-(i), B-(iv), C-(ii), D-(iii)

(4) A-(iv), B-(ii), C-(i), D-(iii)

24. The diagram below shows restriction enzyme sites in the E.coli cloning VECTOR pBR322 determine the function of the gene marked as P and Q.



(1) The GENE 'P' provides resistance to antibiotic while 'Q' is involved in the replication of the plasmid.

(2) The GENE 'P' is required for initiating replication and 'Q' code for the protein involved in the replication of plasmid.

(3) 'P' is responsible for restriction site recognition and 'Q' for resistance to antibiotics.

(4) 'P' is responsible for maintaining plasmid stability and 'Q' for resistance to antibiotic.

25. What does "Ti" denote in the plasmid of *Agrobacterium tumefaciens* ?

(1) Transcription intiator

(2) Tumor inducer

(3) T-DNA GENE

(4) Tumor inhibitor GENE

26. Identify the correct statement regarding bioreactors :

(1) Bioreactors are primarily used in Laboratories for small experimental cultures.

(2) The majority of industrial bioreactors are air lift type rather than stirred-tank type.

(3) Bioreactor maintain suitable conditions for microbial or cell growth to maximize product formation.

(4) Bioreactors are not equipped with systems to regulate temperature, pH and oxygen level.

27. Match List-I with List-II.

	List-I		List-II
(A)	Comb plates	(I)	Osteicthyes
(B)	Feather like gills	(II)	Hemichordata
(C)	Worm like animal	(III)	Ctenophora
(D)	Operculum	(IV)	Mollusca

Choose the correct answer from the options given below :

(1) A-IV, B-I, C-III, D-I

(2) A-III, B-IV, C-II, D-I

(3) A-II, B-III, C-IV, D-I

(4) A-II, B-III, C-I, D-IV

28. Consider the following statements :
(A) *Hirudinaria* is true coelomate
(B) Liver fluke is pseudocoelomate
(C) Filariaworm is acoelomate
(D) *Pleurobrachia* is pseudocoelomate.
Choose the correct answer from the options given below :

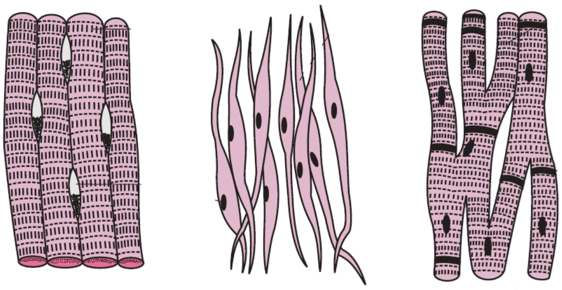
(1) B only (2) A only (3) C only (4) D only

29. The following are the statements about non-chordates :
(A) Pharynx is perforated by gill slits.
(B) Notochord is ventrally located.
(C) Central nervous system is ventral
(D) Post anal tail may be present.
Select the correct combination from the option given below :

(1) Only A is correct (2) A, B & C are correct

(3) Only C is correct (4) C & D are correct

30. Given below the diagram. Identify these muscles (A, B and C) :-



(1) A - Smooth muscle, B - striated muscles, C - cardiac muscle

(2) A - Striated muscle, B - smooth muscle, C - cardiac muscle

(3) A - Involuntary muscle, B - voluntary muscle, C - cardiac muscle

(4) A - Cardiac muscle, B - smooth muscle, C - Striated muscle

31. Match the List-I with List-II :-

List-I		List-II	
(A)	Hing joints	(I)	Flat bone of skull
(B)	Fibrous joints	(II)	Between the carpals
(C)	Gliding joints	(III)	Knee joint
(D)	Pivot joints	(IV)	Between atlas and Axis

Choose the correct answer from the options given below :-

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-I, D-IV

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-I, C-II, D-IV

32. Which of the following is an auto-immune disorder ?

(1) Rheumatoid arthritis

(2) Gout

(3) Muscular dystrophy

(4) Tetany

33. Match List-I with List-II :-

List-I		List-II	
(A)	*Datura*	(I)	*Erythroxylum coca*
(B)	Smack	(II)	*Cannabis sativa*
(C)	Crack	(III)	Hallucinogenic properties
(D)	Hashish	(IV)	*Papaver somniferum*

Choose the correct answer from the options given below :-

(1) A-IV, B-III, C-I, D-II

(2) A-I, B-III, C-II, D-IV

(3) A-II, B-I, C-III, D-IV

(4) A-III, B-IV, C-I, D-II

34. Match List-I with List-II :-

List-I		List-II	
(A)	Pneumonia	(I)	Fungus
(B)	Malaria	(II)	Nematode
(C)	Ringworm	(III)	Protozoa
(D)	Elephantiasis	(IV)	Bacteria

Choose the correct answer from the options given below :-

(1) A-I, B-III, C-II, D-IV (2) A-IV, B-III, C-I, D-II

(3) A-III, B-I, C-IV, D-II (4) A-II, B-IV, C-III, D-I

35. Given below are two statements :-

Statement-I :- Bone marrow is the secondary lymphoid organ where all blood cells including lymphocytes are produced.

Statement-II :- Both bone marrow and thymus provide micro environments for the development and maturation of B-Lymphocytes.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

36. Which of the following statement is **correct** regarding the process of replication in E.coli.

(1) The RNA dependent DNA polymerase catalyse polymerization in one direction 3' ⟶ 5'

(2) The DNA dependent RNA polymerase catalyses polymerization in 5' ⟶ 3' as well as 3' ⟶ 5' direction.

(3) The DNA dependent DNA polymerase catalyse polymerization in 5' ⟶ 3' direction.

(4) The RNA dependent RNA polymerase catalyse polymerization in 3' ⟶ 5' direction

37. Match List-I with List-II.

(A)	Down's syndrome	(I)	16th chromosome
(B)	Kline felter's syndrome	(II)	Trisomy in Autosome
(C)	β thalassemia	(III)	Sex chromosome
(D)	α Thalassemia	(IV)	11^{th} chromosome

(1) A-II, B-III, C-I, D-IV (2) A-II, B-III, C-IV, D-I

(3) A-III, B-II, C-IV, D-I (4) A-I, B-IV, C-II, D-III

38. Which one is the correct product of DNA dependent RNA polymerase to the given template.
3'TACATGGCAAATATCCATTCA5'

(1) 5'TACATGGCAAATATCCATTCA3'

(2) 3'TACAACCGTTTATAGGTAAGT5'

(3) 5'AUGUACCGUUUAUAGGUAAGU3'

(4) 5'ATGTACCGTTTUCAGGTAAGU3'

39. Match List-I with List-II :-

List-I		List-II	
(1)	Splicing of intervening sequence	(I)	Promoter region in DNA
(2)	TATA Box	(II)	Small nuclear Ribonucleo protein
(3)	RNA polymerase III	(III)	Rho factor
(4)	Termination of transcription	(IV)	5s r-RNA

(1) 1-II, 2-I, 3-IV, 4-III (2) 1-I, 2-II, 3-III, 4-IV

(3) 1-II, 2-III, 3-IV, 4-I (4) 1-IV, 2-III, 3-I, 4-II

40. As per ABO blood grouping system. The blood group of father is B^+ and mother is AB^+ and child is B^+. Their respective genotype is :-

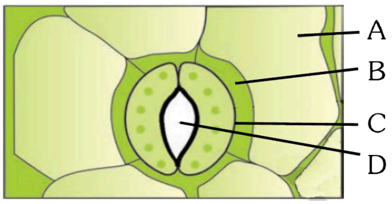
(1) $I^Bi/I^AI^B/I^Bi$

(2) $I^Bi/I^Ai/I^Bi$

(3) $I^AI^B/I^Bi/I^BI^B$

(4) $I^BI^B/I^AI^B/I^A{}_i$

41. In the given figure which component are modified epidermal cell ?



(1) C

(2) D

(3) A

(4) B

42. Bulliform cells are responsible for :-

(1) Maintain pressure gradiant in guard cell.

(2) Increased photosynthesis in dicots.

(3) Inward curling of leaves in monocots.

(4) Opening of stomata

43. Given below are two statements :

Statement-I :- Parenchyma is living mechanical tissue but collenchyma is soft tissue.

Statement-II :- Gymnosperms lack xylem vessels but presence of xylem vessels is the characteristic of angiosperm.

In the light of the above statements, choose the correct answer from the option given below :-

(1) Both statement I and statement II are true.

(2) Both statement I and statement II are false.

(3) Statement I is true and statement II false.

(4) Statement I is false and statement II true

44. Identify the set of *incorrect* statements :-

A.The flowers of *vallisnaria* are colourless and produce fragrence.

B. The Flower of waterlily are pollinated by water.

C. In most of water pollinated species, the pollen grain not covered by mucilagenous layer.

D. Pollengrain of some hydrophyte are long end ribbon like.

E. In most of hydrophyte, the pollen grains are carried passively inside water.

Choose the correct answer from the option given below :-

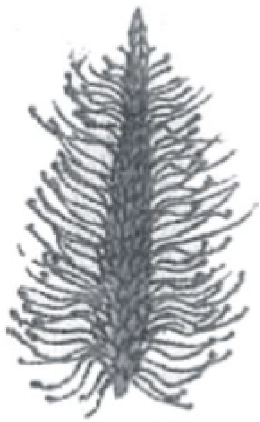
(1) C,D and E only

(2) A,B,C and D only

(3) A,C,D,E only

(4) A,B,C and E only

45. Identify the correct description about the given figure :-



(1) Wind pollinated plant inflorescence showing flowers with well exposed stamens.

(2) Wind pollinated Flower showing stamen with mucilagenous covering.

(3) Cleistogamous flowers showing Xenogamy

(4) Compact inflorescence showing only geitonogamy

46. Match the list-I with list-II

List-I		List-II	
(A)	Non-medicated IUD	(I)	Make the uterus unsuitable for implantation
(B)	Copper releasing IUD	(II)	Block gamete transport
(C)	Hormone releasing IUD	(III)	Suppress sperm motility
(D)	Sterlisation	(IV)	Phagocytosis of sperm

(1) A-III, B-IV, C-II, D-I

(2) A-IV, B-II, C-III, D-I

(3) A-II, B-III, C-I, D-IV

(4) A-IV, B-III, C-I, D-II

47. Which of the following is a Natural/traditional contraceptive method ?

(1) Condom

(2) Cu-T

(3) Vaults

(4) Periodic abstinence

48. Given below are two statements

Statement-I :- The uterus open into vagina through narrow cervix.

Statement-II :- Cervix along with vagina form a birth canal.

In the light of the above statements choose the correct answer from the option given below.

(1) Both Statement-I and Statement-II are true

(2) Both Statement-I and Statement-II are false

(3) Statement-I is true but Statement-II is false

(4) Statement-I is false but Statement-II is true.

49. Which of the following is not a component of vulva ?

(1) Penile urethra (2) Labia Majora

(3) Labia Minora (4) Clitoris

50. Given below are two statements one is labelled as Assertion (A) and the other is labelled as a Reason (R)

Assertion (A) :- The mammary glands of the female undergo differentiation during Pregnancy

Reason (R) :- Level of Progesterone hormone are very high during Pregnancy.

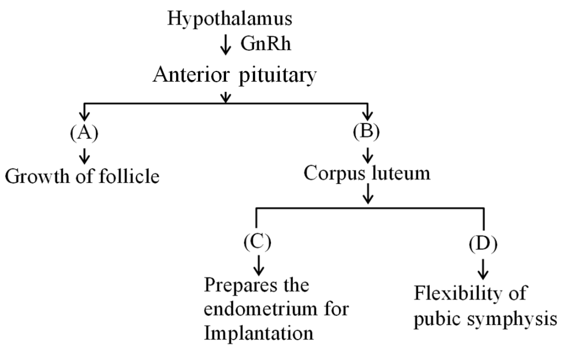
(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is not a correct explanation of A.

(3) A is correct but R is not correct

(4) A is not correct but R is correct

51. Identify the correct option (A),(B),(C),(D) with respect to role of Hormones in female.



(1) FSH, LH, Progesterone, Relaxin

(2) FSH, LH, Progesterone, oestrogen

(3) FSH, LH, Progesterone, oxytocin

(4) FSH, LH, oestrogen, Relaxin

52. Two snapdragon plants are crossed and it result equal number of Red & Pink flower plants. Select the correct combination of parents.

(1) Red & Pink flower plant

(2) Both Red flower

(3) Red & white flower

(4) Both Pink flower plant

53. In lac operon the repressor is inactivated by interaction with ?

(1) Inducer (2) Glucose

(3) RNA polymerase (4) Galactose

54. If a impure voilet & white flowered pea plant is crossed, then what would be the ratio of voilet & white flower plants in progeny :-

(1) 3 : 1 (2) 1 : 3 (3) 1 : 2 : 1 (4) 1 : 1

55. Which site is to be located towards 5' end or upstream of the structural gene (in reference of polarity of coding strand) :-

(1) Terminator (2) Promotor

(3) Operator (4) Repressor gene

56. Which statements do **not** support the Mendel's law of dominance ?

(A) Factors occur in pairs.

(B) In F_1 both alleles express themselves.

(C) In F_1 none of the alleles is completely dominant.

(D) Segragation of one pair of character is independent of other pair of character.

(E) Characters are controlled by discrete units called factor.

Choose the correct answer from the options given below.

(1) A, B, C, D, E (2) A and E only

(3) A, B and C only (4) B, C and D only

57. Lecithin, found in cell membrane, is an example of : -

(1) Proteins

(2) Polysaccharides

(3) Glycerol

(4) Phospholipid

58. When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme is a classical example of :

(1) Feed back inhibition

(2) Competitive inhibition

(3) Cofactor inhibition

(4) Enzyme activation

59. Match List-I with List-II.

	List-I		List-II
(A)	Antibody	(I)	Inter cellular ground substance
(B)	Receptor	(II)	Fights with infectious agents
(C)	Trypsin	(III)	Sensory reception
(D)	Collagen	(IV)	Enzyme

Choose the correct answer from the options given below :

(1) A-IV, B-I, C-II, D-III
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-III, C-IV, D-I
(4) A-III, B-IV, C-I, D-II

60. Which of the following show the correct order of structures involved in generation and propogation of an action potential in the human heart ?
(A) AV Bundle
(B) Atrio - ventricular node
(C) Sino - atrial node
(D) Purkinge fibres
(E) Right and left bundle branches
Choose the correct sequence.

(1) $C \rightarrow B \rightarrow A \rightarrow E \rightarrow D$
(2) $B \rightarrow C \rightarrow A \rightarrow D \rightarrow E$
(3) $C \rightarrow B \rightarrow E \rightarrow A \rightarrow D$
(4) $C \rightarrow A \rightarrow B \rightarrow E \rightarrow D$

61. Match List-I with List-II :-

List-I		List-II	
(A)	P-wave	(I)	Electrical inactivity in heart muscles
(B)	QRS wave complex	(II)	Electrical excitation of ventricles
(C)	End of T wave	(III)	End of systole
(D)	Interval between T and next P	(IV)	Electrical excitation of atrial muscles

Choose the correct answer from the option given below-

Options :-

	A	B	C	D
(1)	I	II	III	IV
(2)	IV	II	III	I
(3)	IV	III	II	I
(4)	IV	I	II	III

62. **Statement-I :-** The cerebrum wraps around a structure called thalamus. Which is a major coordinating centre for sensory and motor signaling.
Statement-II :- Cardiovascular reflexes are controlled by the medulla oblongata.

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect.
(4) **Statement I** is incorrect but **Statement II** is correct.

63. Eyes of an octopus and of mammals are examples of the -

(1) Adaptive radiation
(2) Natural selection
(3) Convergent evolution
(4) Divergent evolution

64. Given below are some stages of human evolution. Arrange them in correct sequence (recent to past)

(A) *Homo habilis*
(B) *Homo sapiens*
(C) *Homo neanderthalensis*
(D) *Homo erectus*
Choose the correct sequence of human evolution from the options given below :

(1) D-A-C-B
(2) B-A-D-C
(3) A-D-C-B
(4) B-C-D-A

65. Which of the following factors will affect the Hardy-weinberg equilibrium ?

(1) Large population
(2) Random mating
(3) Constant gene pool
(4) Mutation

66. Match List-I with List-II.

	List-I		List-II
(A)	Proterozoic era	(i)	Age of dinosaurs
(B)	Paleozoic era	(ii)	Age of fishes and amphibia
(C)	Mesozoic era	(iii)	Age of angiosperms and mammals
(D)	Cenozoic era	(iv)	Lower invertebrate

Choose the correct answer from the options given below :

(1) A-i, B-iii, C-iv, D-ii

(2) A-ii, B-i, C-iii, D-iv

(3) A-iii, B-iv, C-i, D-ii

(4) A-iv, B-ii, C-i, D-iii

67. Match List-I with List-II.

	List-I		List-II
(A)	Fermentation	(i)	On inner mitochondrial membrane
(B)	Link-reaction	(ii)	Mobile electron carrier
(C)	Breakdown of succinate	(iii)	Matrix
(D)	Cytochrome c	(iv)	Cytoplasm

Choose the correct answer from the options given below :

(1) A-iv, B-iii, C-ii, D-i (2) A-i, B-ii, C-iii, D-iv

(3) A-iv, B-iii, C-i, D-ii (4) A-iii, B-i, C-iv, D-ii

68. Given below are two statements :

Statement-I : In C_3-plants, some O_2 binds to PEPcase hence CO_2 fixation is decreased.

Statement-II : In C_3-plants, bundle sheath cells show photorespiration.

In the light of the above statements, choose the correct answer from the option given below :

(1) Both Statements are true.

(2) Both Statements are false.

(3) Statement-I is true but Statement-II is false.

(4) Statement -I is false but Statement-II is true.

69. Spraying juvenile conifers with which of the following plant growth regulators to promote early seed production ?

(1) Cytokinin (2) Ethylene

(3) Auxin (4) Gibberellin

70. Which one of the following is non-motile sexual spore in fungi :

(1) Aplanospore (2) Zoospore

(3) Ascospore (4) Both 1 and 3

71. Match List-I with List-II.

	List-I		List-II
(A)	*Mucor*	(i)	Edible fungi
(B)	*Albugo*	(ii)	Imperfect fungi
(C)	*Truffles*	(iii)	Bread mould
(D)	*Trichoderma*	(iv)	Parasitic fungi on mustard

Choose the correct answer from the options given below :

(1) A-iii, B-iv, C-i, D-ii

(2) A-i, B-ii, C-iii, D-iv

(3) A-iii, B-iv, C-ii, D-i

(4) A-ii, B-iii, C-iv, D-i

72. Read the following statements regarding the member's of ascomycetes.

(A) Motile asexual spores are produced exogenously.

(B) Immotile sexual spores are produced endogenously.

(C) Edible members are commonly known as mushroom.

(D) Mycelium is branched and aseptate.

Choose the incorrect answer from the options given below :

(1) B only (2) B and D only

(3) A, B and C only (4) A, C and D only

73. How many species according to IUCN red list 2004 are documented as extinct species in the last 500 years ?

(1) 784 (2) 338 (3) 359 (4) 87

74. Which is related to latitude gradient in species diversity :-

(A) Columbia has nearly 1400 species of bird while New York at 41°N has 105 species of bird.
(B) New York at 71°N has 105 species of bird while Green land at 41°N has 56 species of bird.
(C) Amazon rain forest have more mammal species than amphibian species.
(D) Tropics harbour more species than temperate.
(E) India has more species of bird than Green land.

Choose the correct answer from the options given below.

(1) A, B, C & E
(2) B, C & D
(3) C, D & E only
(4) Only A, D & E

75. In the equation of Verhulst pearl logistic growth is given below :-

$$\frac{dN}{dt} = rN\left[\frac{K-N}{K}\right]$$

From this equation 'r' indicates.

(1) Biotic potential
(2) Carrying capacity
(3) Environmental resistance
(4) Intrinsic rate of natural increase

76. In India ecologically unique and Biodiversity-rich regions are legally protected as biosphere reserve, national park, and sanctuaries. Which type of conservation is it.

(1) In Vitro conservation
(2) Cultural conservation
(3) Heritage conservation
(4) In-situ conservation

77. Which of the following are not required for the dark reaction of photosynthesis ?

(A) Light
(B) Chlorophyll
(C) CO_2
(D) NADPH
(E) ATP

Choose the correct answer from the options given below :

(1) C, D and E only
(2) A, C and E only
(3) A and B only
(4) B and E only

78. Why decapitation is widely applied in tea plantation and hedge making ?

(1) Inhibits the growth of lateral buds.
(2) Promotes the growth of lateral buds.
(3) Inhibits the growth of lateral shoots.
(4) Promotes the apical dominance.

79. Formation of tracheary elements from the cells of shoot apical meristems is an example for

(1) Differentiation
(2) Redifferentiation
(3) Dedifferentiation
(4) Elongation

80. How many molecules of ATP and NADPH are required in reduction for every molecule of CO_2 fixed in the calvin cycle ?

(1) 2 molecules of ATP and 3 molecules of NADPH
(2) 2 molecules of ATP and 2 molecules of NADPH
(3) 3 molecules of ATP and 3 molecules of NADPH
(4) 3 molecules of ATP and 2 molecules of NADPH

81. Identify the step in aerobic respiration, which does not involve decarboxylation of substrate ?

(1) Citric acid $\longrightarrow$ α-ketoglutaric acid
(2) Pyruvic acid $\longrightarrow$ acetyl-CoA
(3) Succinic acid $\longrightarrow$ malic acid
(4) α-ketoglutaric acid $\longrightarrow$ succinic acid

82. Which bonds are formed between metal ions and side chains at active site of enzymes ?

(1) Ionic bonds
(2) Covalent bonds
(3) Coordination bonds
(4) Peptide Bonds

83. Match the List-I with List-II related to digestive system of cockroach.

	List-I		List-II
(A)	Structures used for grinding of food	(I)	Pharynx
(B)	Located at the junction of foregut and midgut	(II)	Hepatic caeca
(C)	Located at the junction of midgut and hindgut	(III)	Malpighian tubules
(D)	Mouth opens into	(IV)	Mandibles

(1) A-IV, B-II, C-III, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-III, B-II, C-IV, D-I

84. In male cockroach a pair of unjointed filamentous structure called anal style are present on

(1) 5^{th} segment (2) 9^{th} segment

(3) 8^{th} and 9^{th} segment (4) 11^{th} segment

85. Which of the following factors are favourable for dissociation of O_2 from oxyhaemoglobin in tissues ?

(1) High pO_2 and High pCO_2

(2) High pCO_2 and More H^+ concentration

(3) Low pCO_2 and High H^+ concentration

(4) Low pCO_2 and Low temperature

86. Match List-I with List-II.

	List-I		List-II
(A)	Tidal volume (TV)	(I)	1000-1100 ml
(B)	Inspiratory reserve volume (IRV)	(II)	500 ml
(C)	Expiratory reserve volume (ERV)	(III)	1100-1200 ml
(D)	Residual volume (RV)	(IV)	2500-3000 ml

(1) A-II, B-IV, C-I, D-III (2) A-III, B-II, C-IV, D-I

(3) A-II, B-I, C-IV, D-III (4) A-I, B-III, C-II, D-IV

87. Which of the following is a steroid hormone?

(1) Insulin (2) Glucagon

(3) Cortisol (4) Epinephrine

88. **Assertion (A) :** Leydig cells secrete gonadotrophins in male while growing ovarian follicles secrete estrogen in female.

Reason (R) : FSH acts upon ovarian follicles in females to stimulates their growth and development.

With the respect of above statements choose the correct answer from the options given below -

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true and R is not correct explanation of A.

(3) A is true and R is false

(4) A is false and R is true.

89. Match list-I with list-II -

	List-I		List-I
A.	Excessive secretion of growth hormone	I	Exopthalmic goitre
B.	Hyperthyroidism, characterised by enlargement of thyroid gland	II	Severe disfigurement, especially in face
C.	Hypothyroidism, stunted growth and mental retardation	III	Cushing's syndrome
D.	Hypersecretion of cortisol, Red moon face, increase in blood glucose	IV	Cretinism

Choose the correct answer from the options given below -

(1) A-I, B-II, C-IV,D-III

(2) A-II, B-I, C-III,D-IV

(3) A-II, B-I, C-IV,D-III

(4) A-III, B-I, C-IV,D-II

90. Choose the correct sequence for the enzyme action :-

(1) $E + S \longrightarrow$ ES complex $\rightleftarrows$ EP complex $\longrightarrow E + P$

(2) $E + S \rightleftarrows$ ES complex $\rightleftarrows$ EP complex $\longrightarrow E + P$

(3) $E + P \rightleftarrows$ EP complex $\longrightarrow$ ES complex $\longrightarrow E + S$

(4) $E + S \rightleftarrows$ ES complex $\longrightarrow$ EP complex $\longrightarrow E + P$

SUBJECT : PHYSICS

Topic : FULL SYLLABUS.

91. A bob is whirled in a horizontal plane by means of a string with a speed v. If length of string becomes 3 times while keeping the tension same, the speed of bob becomes-

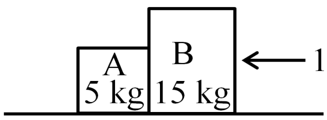
(1) $\sqrt{2}$ v (2) $\sqrt{3}$ v (3) $\frac{v}{\sqrt{3}}$ (4) 3v

92. In a vernier callipers, (N) divisions of vernier scale coincide with (N-2) divisions of main scale. If 1 MSD represents 0.05 mm, the vernier constant (in cm) is :-

(1) $\frac{1}{100N}$ (2) $\frac{1}{100(N-1)}$

(3) 100N (4) 100(N−1)

93. A horizontal force of 100N is applied to a block B as shown in figure. The mass of blocks A and B are 5 kg and 15 kg. The blocks slide over a frictionless surface. The force exerted by block A on block B is-



(1) Zero (2) 25 N

(3) 75 N (4) 50 N

94. The maximum elongation of a copper wire of 10 m length if the elastic limit of copper and its young's modulus respectively are 1.5×10^8 N/m^2 and 12×10^{10} N/m^2 is-

(1) 1.25 cm (2) 12.5 cm

(3) 125 cm (4) 12.5 m

95. At any instant of time t, the displacement of a particle is given by $(1-6t)\hat{i}$ (in SI unit) under the influence of force of $2\hat{i}$ (SI unit). The value of instantaneous power is (SI unit) :-

(1) 12 (2) −12

(3) −6 (4) 6

96. Two bodies P and Q of mass m and 2m respectively, undergo completely inelastic one dimensional head-on collision. Initially the body P moves with velocity v_1 while body Q moves with velocity $\frac{v_1}{2}$ in the same direction. The velocity of system after collision is v_2. The ratio $v_1 : v_2$ is-

(1) 1 : 2 (2) 2 : 1 (3) 2 : 3 (4) 3 : 2

97. The moment of inertia of a thin rod about an axis passing through one of its ends and perpendicular to the rod is 27 kg-m^2. The length of this 1 kg rod is-

(1) 3m (2) 6m (3) 9m (4) 12m

98. The quantity which have the same dimensions as those of surface tension:-

(1) Coefficient of viscosity

(2) Spring constant

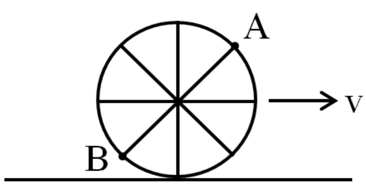
(3) Force

(4) Impulse

99. A square wire frame of side 10 cm is placed gently over the surface of water. If surface tension of water is 0.07 N/m, then the excess force required to take it away from the surface is-

(1) 5.6 mN (2) 56 mN (3) 28 mN (4) 2.8 mN

100. A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown which one of the following options is correct-



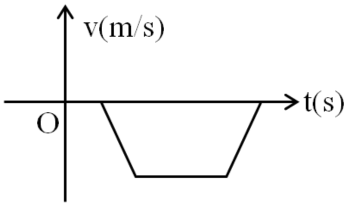
(1) Point A moves slower than point B.

(2) Point A moves faster than point B.

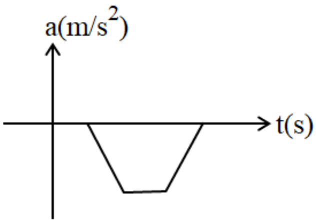
(3) Both the point A and B moves with equal speed.

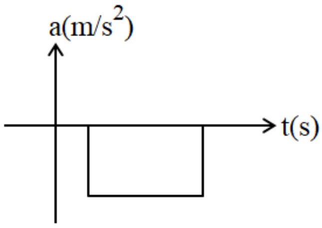
(4) Point B has zero speed.

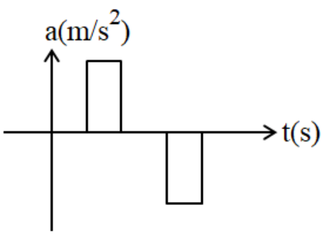
101. The velocity (v) - time (t) plot of the motion of a body is shown below-

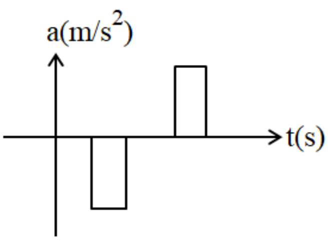


The acceleration (a) - time (t) graph that best suits this motion is-

(1)


(2)


(3)


(4)


102. The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R, in a circular orbit at an altitude of 3R from the surface of the earth is-

(1) $\frac{2GmM}{3R}$ (2) $\frac{GMm}{8R}$

(3) $\frac{7GMm}{8R}$ (4) $\frac{5GmM}{6R}$

103. In a series LCR circuit, the inductance L is 20 mH, capacitance $2\mu F$ and resistance R is 100Ω. The frequency on which resonance occurs is-

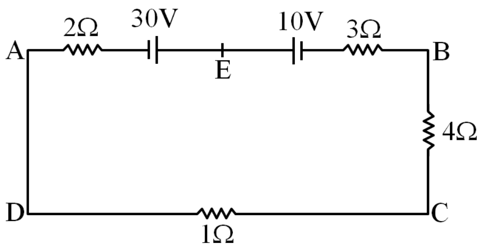
(1) 0.9 KHz

(2) 1.2 KHz

(3) 0.8 KHz

(4) 2 KHz

104. The magnitude and direction of the current in the following circuit is-



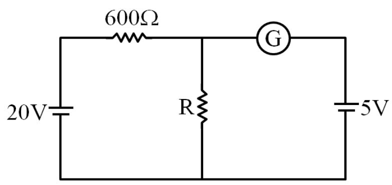
(1) 0.2 A from A to B through E.

(2) 2 A from A to B through E.

(3) 2 A from B to A through E.

(4) 0.2 A from B to A through E.

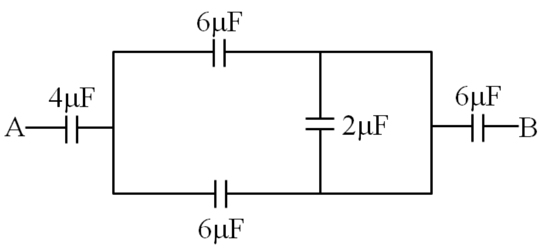
105. If the galvanometer G does not show any deflection in the circuit then the value of R is given by-



(1) 200Ω (2) 100Ω

(3) 300Ω (4) 400Ω

106. The equivalent capacitance between A and B of the system shown in the following circuit is-



(1) 2 μF (2) 4 μF (3) 6 μF (4) 8 μF

107. A capacitor has a capacitance of $100\mu F$ and it is connected to a voltage source that changes at a rate of 20 V/s. What is the displacement current?

(1) 1 mA (2) 2 mA

(3) 3 mA (4) 4 mA

108. In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2×10^{10} Hz and amplitude 24 V/m. Then the amplitude of oscillating magnetic field is (Speed of light in free space = 3×10^8 m/s)

(1) 8×10^{-8} T

(2) 0.8×10^{-8} T

(3) 8×10^{-7} T

(4) 0.8×10^{-7} T

109. A wire carrying a current 5A along the positive x-axis has length 2m. It is kept in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 4\hat{k})T$. The magnitude of the magnetic force acting on the wire is-

(1) 50N (2) 30N

(3) 20N (4) 40N

110. The resistance of platinum wire at 0ºC is 4Ω and 8.4Ω at 60ºC. The temperature coefficient of resistance of the wire is-

(1) $0.183°C^{-1}$

(2) $1.83°C^{-1}$

(3) $18.3°C^{-1}$

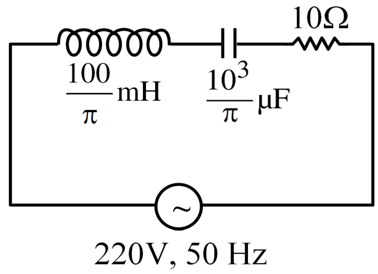
(4) $0.0183°C^{-1}$

111. 20 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. When these are connected in parallel to the same battery, the current become n times. The value of n is :-

(1) 400 (2) 200

(3) 100 (4) 500

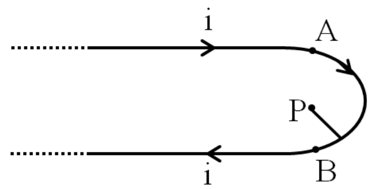
112. The net impedance of circuit (as shown in figure) will be-



(1) 10Ω (2) 20Ω

(3) 50Ω (4) 30Ω

113. A very long conducting wire is bent in a semicircular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by-



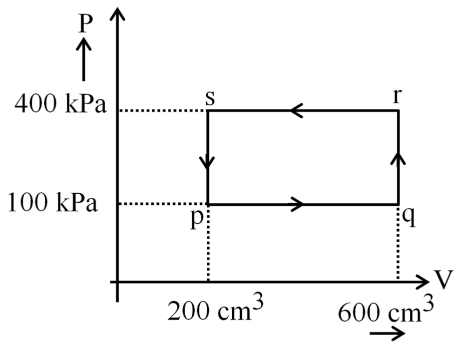
(1) $\frac{\mu_0 I}{2R}$ pointed outward from the page

(2) $\frac{\mu_0 I}{4R}\left[1+\frac{2}{\pi}\right]$ pointed outward from the page

(3) $\frac{\mu_0 I}{4R}\left[1+\frac{2}{\pi}\right]$ pointed inward to the page

(4) $\frac{\mu_0 I}{4R}$ pointed inward to the page

114. A thermodynamic system is taken through the cycle pqrsp. The work done by gas along the path rs is-



(1) Zero (2) –50J

(3) –160J (4) 80J

115. A metallic bar of Young's modulus, 1×10^{11} Nm^{-2} and coefficient of linear thermal expansion 10^{-6} ºC^{-1}, length 2m and area of cross section 10^{-4} m^2 is heated from 0ºC to 100ºC without expansion or bending. The compressive force developed in it is-

(1) 1×10^3 N

(2) 4×10^3 N

(3) 200×10^3 N

(4) 0.5×10^3 N

116. The temperature of a gas is –150ºC. To what temperature the gas should be heated so that rms speed of molecules is increased by 2 times?

(1) –450ºC

(2) 1107K

(3) 553ºC

(4) 1887K

117. If $x = 20 \sin (4\pi t + \frac{\pi}{4})$ m represents the motion of a particle executing simple harmonic motion, then time period and amplitude of motion respectively are-

(1) $\frac{1}{2}$s, 20 cm

(2) 2s, 20 m

(3) $\frac{1}{2}$s, 20 m

(4) 2s, 40 cm

118. The ratio of frequencies of first overtones produced by an open organ pipe to that of closed organ pipe having the same length is-

(1) 1 : 3

(2) 4 : 3

(3) 5 : 2

(4) 2 : 3

119. If the mass of the bob in a simple pendulum is reduced to half its original mass and its length is made thrice its original length, then the new time period of oscillation is $\frac{x}{3}$ times its original time period. Then the value of x is-

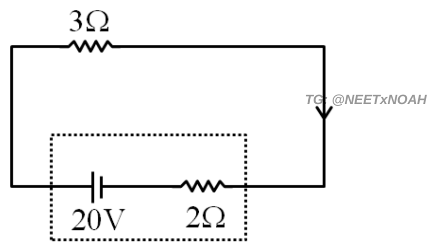
(1) $\frac{1}{\sqrt{3}}$

(2) $2\sqrt{3}$

(3) $3\sqrt{3}$

(4) $\sqrt{3}$

120. The terminal voltage of the battery, whose emf is 20V and internal resistance 2Ω, when connected through an external resistance of 3Ω as shown in the figure.



(1) 10V

(2) 20V

(3) 12V

(4) 16V

121. **Assertion :** Net electric field inside conductor is zero in electrostatic condition.

Reason : Potential inside conductor is same at all point inside it.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

122. Match List-I with List-II :

	List-I (Material)		List-II Susceptibility (χ)
(A)	Diamagnetic	(I)	$\chi \propto \frac{1}{T}$
(B)	Ferromagnetic	(II)	$\chi \propto T^0$
(C)	Paramagnetic	(III)	$\chi \propto \frac{1}{T - T_c}$
(D)	Non-Magnetic	(IV)	$\chi = 0$

Choose the correct answer from the options given below-

(1) (A–I), (B–IV), (C–IV), (D–II)

(2) (A–II), (B–III), (C–I), (D–IV)

(3) (A–III), (B–I), (C–IV), (D–II)

(4) (A–IV), (B–II), (C–I), (D–III)

123. A bullet is fired from a gun in the direction 45° above the horizontal. If maximum height attained by the bullet is 1 km, find projection speed of bullet. ($g = 10\ m/s^2$)

(1) 280 m/s (2) 200 m/s

(3) 400 m/s (4) $100\sqrt{2}$ m/s

124. Two bodies of mass 4m and 16m are placed at a distance R. The gravitational potential on the line joining the bodies where the gravitation field equals zero, will be (G = Gravitational constant) :-

(1) $\frac{-12Gm}{R}$ (2) $\frac{-24Gm}{R}$

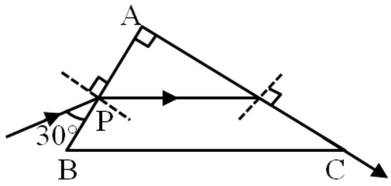
(3) $\frac{-36Gm}{R}$ (4) $\frac{-48Gm}{R}$

125. A horizontal bridge of height 100 m is built across a river. A student standing on the bridge throws a small ball vertically downwards with a speed u. The ball strikes the water surface after 2 sec. Find value of u. ($g = 10\ m/s^2$)

(1) 20 m/s (2) 40 m/s

(3) 60 m/s (4) 80 m/s

126. A light ray enters a right angled prism at point P as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. Find the refractive index of prism.



(1) $\frac{\sqrt{5}}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{\sqrt{7}}{2}$ (4) $\frac{\sqrt{9}}{2}$

127. A light ray travels distance 's' in time t_2 in a denser medium and distance '10s' in a rarer medium in time t_1. If the critical angle of denser medium with respect to the rarer medium is 30°; find ratio $\frac{t_1}{t_2}$.

(1) 1 : 20 (2) 10 : 1 (3) 5 : 1 (4) 1 : 2

128. For Young's double slit experiment, two statements are given below :-

Statement-I :- If distance between the slits is increased, the fringe width increases.

Statement-II :- If the sources of light used are of green and red light alternatively, the angular fringe width will be greater for green light as compared to that for red light.

(1) Both Statement I and Statement II are true.

(2) Statement I is true but Statement II is false.

(3) Statement I is false but Statement II is true.

(4) Both Statement I and Statement II are false.

129. An unpolarised light beam travelling in air, strikes a glass surface at Brewster's angle; then which of the following statements are correct ?

(a) The refracted light will be partially polarised

(b) The reflected light will be completely polarized.

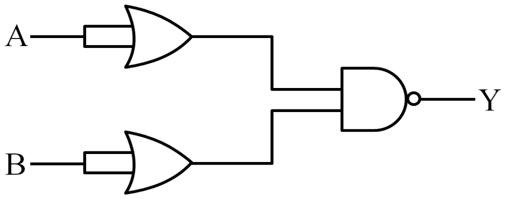
(c) The refractive index of glass with respect to air will be equal to 'tangent of angle of incidence'.

(d) Both the reflected and refracted light will be partially polarised

(1) c and d only (2) a and b only

(3) a, b and c only (4) a, b, c and d

130. The following circuit comprises different logic gates as shown in diagram :-



For inputs A and B, the output (Y) of the circuit is similar to the output of :-

(1) NAND gate

(2) AND gate

(3) XOR gate

(4) NOT gate

131. $X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} {}_{81}Q^{286}$

In the nuclear emission stated above the mass number and atomic number of X respectively are :-

(1) 286, 81

(2) 290, 80

(3) 288, 82

(4) 290, 82

132. The correct statements about photon among the following are : (Here C is the velocity of light and ν is the frequency of light)

(A) The energy of photon is $E = \frac{hc}{\lambda}$

(B) The momentum of a photon, $P = \frac{h\nu}{C}$

(C) In a photon-electron collision total energy is conserved but total momentum is not conserved

(D) Photon possesses negative charge.

Choose the correct answer from the options given below :-

(1) A and B only

(2) A, B, C and D all

(3) A, B and C only

(4) A, B and D only

133. In hydrogen spectrum the shortest wavelength in the Balmer series is λ. The shortest wavelength in the Pfund series is :-

(1) 4λ

(2) $\frac{25}{4}\lambda$

(3) $\frac{4\lambda}{25}$

(4) 16λ

134. If incident electromagnetic radiation has an incident energy of 2.40 eV, the work functions of Caesium (Cs), Potassium (K) and sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively which of these photosensitive surfaces may emit photoelectrons ?

(1) Both Cs and K

(2) K only

(3) Na only

(4) Cs only

135. Given below are two statements :-

Statement I :- When zener diode is used as DC voltage regulator, then with the increase in input voltage V_i, zener voltage V_z remains unchanged. But when V_i decreases, then V_z also decreases.

Statement II :- LED's requires low operational voltage and less power as compare to incandescent lamp.

In the light of the above statements, choose the most appropriate answer from the options given below :-

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

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136. The method used for quantitative estimation of nitrogen is -

(1) Kjeldahl's method (2) Duma's method

(3) Carius method (4) Both 1 & 2

137. Match the Column.

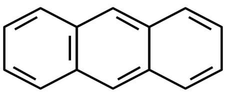
	Column-I (Vitamins)		Column-II (Chemical name)
(A)	Vitamin B_1	(I)	Thiamine
(B)	Vitamin B_6	(II)	Pyridoxine
(C)	Vitamin C	(III)	Ascorbic acid
(D)	Vitamin B_2	(IV)	Riboflavin

(1) A-I, B-II, C-III, D-IV

(2) A-I, B-III, C-II, D-IV

(3) A-I, B-IV, C-III, D-II

(4) A-I, B-IV, C-II, D-III

138. Anthracene  is an aromatic compound which contains (4n + 2) π electrons, value of n will be

(1) 1 (2) 2

(3) 3 (4) 0

139. **Statement-I** : Isobutene on treatment with diborane gives an addition product with formula $\left(\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>CH{-}CH_2\right)_3B$

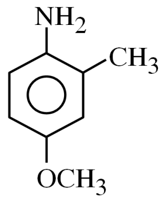
Statement-II : Oxidation $\left(\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>CH{-}CH_2\right)_3B$ with H_2O_2 in presence of alkali gives isobutyl alcohol.

(1) Statement-I is correct but Statement-II is incorrect.

(2) Statement-I is incorrect but Statement-II is correct.

(3) Both Statement-I & Statement-II are correct.

(4) Both Statement-I & Statement-II are incorrect.

140. The correct IUPAC name of  is

(1) 2-methyl-4-methoxy aniline

(2) 4-methoxy-2-methyl aniline

(3) 4-amino-3-methyl anisol

(4) 2-amino-5-methoxy toluene

141. $CH_2 = CH_2 \xrightarrow{H_2O/H^+} A \xrightarrow[\Delta]{I_2/OH^\ominus} CHI_3 + B(\text{Salt})$

B is -

(1) $H - COO^\ominus$ (2) $CH_3 - COO^\ominus$

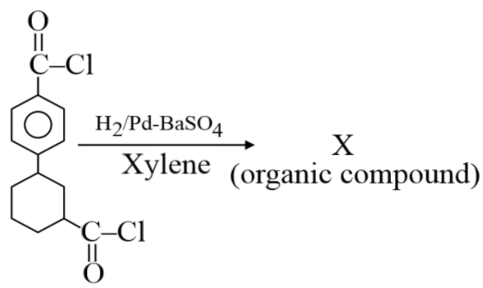
(3) $CH_3 - CH_2 - COO^\ominus$ (4) 1 & 2 both

142. $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} X + H_2O$

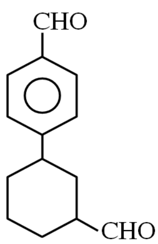
X is -

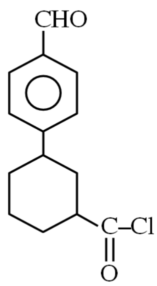
(1) $CH_3{-}OH$ (2) $H{-}CHO$

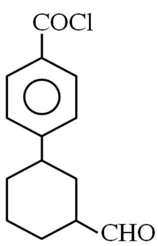
(3) CO_2 (4) CO

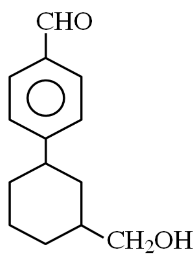
143. 

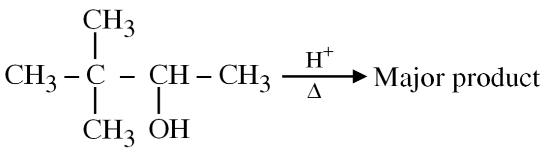
X is -

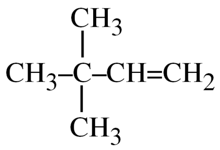
(1) 

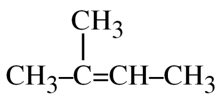
(2) 

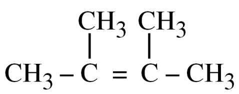
(3) 

(4) 

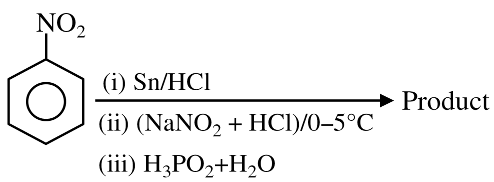
144. 

(1) 

(2) 

(3) 

(4) None of these

145. 

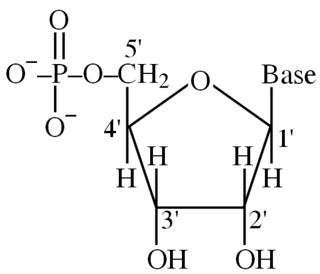
Correct statement about the product -

(1) Compound has three double bond equivalent

(2) Compound can give Fridel-craft reaction

(3) It gives phenol while treating with warm H_2O

(4) It gives carbyl amine reaction

146. 

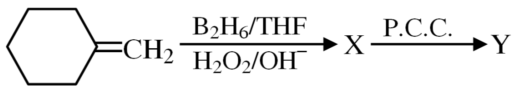
(1) Given unit is nucleoside

(2) Given unit is nucleotide

(3) given unit is dinucleotide

(4) Both 1 & 2 are correct

147. For Y -



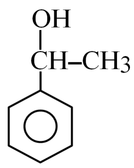
Find incorrect statement -

(1) It reduces fahling solution

(2) It reduces tollen's reagent

(3) It can forms cyanohydrin

(4) It does not react with $NaHSO_3$

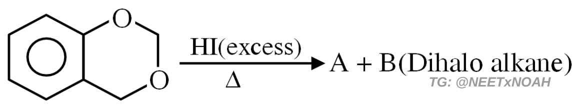
148. 

(1) Given compound belongs to phenol series.

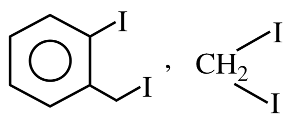
(2) Given compound is aliphatic alcohol.

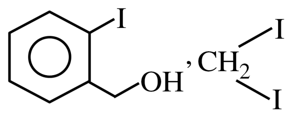
(3) Given compound is aromatic alcohol.

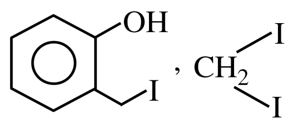
(4) It is 2°-benzylic alcohol.

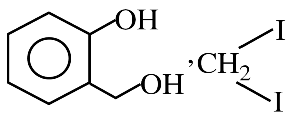
149. 

A & B are -

(1) 

(2) 

(3) 

(4) 

150. $CH_3-CH_2-OH \xrightarrow{PCC} X \xrightarrow{OH^-/\Delta} Y \xrightarrow{LiAlH_4} Z$

Z is -

(1) $CH_3-CH_2-CH_2-CH_2-OH$

(2) $CH_3-CH=CH-CH_2OH$

(3) $CH_3-CH_2-CH_2-CHO$

(4) $CH_3-CH=CH-COOH$

151. The energy of an electron of Li^{+2} in first excited state is XJ then find minimum energy of an electron present in Be^{+3}.

(1) $\frac{64X}{9}$

(2) $\frac{9X}{64}$

(3) X

(4) $\frac{16X}{9}$

152. **Statement-I :** The energy of electron in Be^{+3} ion in n = 4 state is equal to energy of H-atom in n = 1 state.

Statement-II : Third separation energy of electron in Be^{+3} is equal to ionisation energy of an electron of H-atom.

(1) Both statement-I and statement-II are false.

(2) Both statement-I and statement-II are true.

(3) Statement-I is true and statement-II is false.

(4) Statement-I is false and statement-II is true.

153. Atomic weight of X in molecule (X_2) whose 3 moles weigh 150 g.

(1) 50

(2) 25

(3) 100

(4) 75

154. Assign the sign of work done in the following chemical reactions -

(a) $N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$

(b) $NH_4HS(s) \longrightarrow NH_{3(g)} + H_2S_{(g)}$

(c) $C_2H_6(g) + \frac{7}{2}O_{2(g)} \longrightarrow 2CO_{2(g)} + 3H_2O_{(\ell)}$

(1) +, –, –

(2) –, +, –

(3) +, –, –

(4) +, –, +

155. The standard Enthalpy of neutralization is

–13.7 K cal/g eq as

$$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$$

The change in enthalpy during neutralization of 0.75 mol H_2SO_4 by 1.5 mol NaOH.

(1) –20.55 KCal

(2) 27.2 KCal

(3) –6.7 KCal

(4) –54.8 KCal

156. The term $-\frac{dx}{dt}$ in the rate expression refers to-

(1) Average rate of reaction

(2) Instantaneous rate of reaction

(3) Increase in the concentration of reactant

(4) None of these

157. Match List-I with List-II.

List-I		List-II	
(A)	Oxide	(I)	-1
(B)	Super Oxide	(II)	$-\frac{1}{3}$
(C)	Ozonide	(III)	-2
(D)	Peroxide	(IV)	$-\frac{1}{2}$

Choose the correct answer from the option given below :-

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-IV, C-II, D-I

(4) A-III, B-IV, C-I, D-II

158. Find the temperature of the given reaction at equilibrium if $\frac{Kp}{Kc} = 3$.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

(1) 0.27 K

(2) 36.59 K

(3) 0.36 K

(4) 27.3 K

159. In which of the following process Entropy decreases?

(a) $H_2O_{(S)} \longrightarrow H_2O_{(\ell)}$

(b) Polymerisation

(c) Dissolution of salt

(d) Stretching of rubber band

(1) a, b

(2) b, d

(3) b, c, d

(4) a, b, c, d

160. If a graph is plotted between ℓnk v/s $\frac{1}{T}$ for a chemical reaction then value of slope will be-

(1) $\frac{-Ea}{2.303R}$

(2) $\frac{-Ea}{R}$

(3) $\frac{Ea}{R}$

(4) $\frac{Ea}{2.303R}$

161. When 0.2 mol of FeO is oxidised into Fe_2O_3 than quantity of electricity required will be -

(1) 2 F

(2) 0.2 F

(3) 1 F

(4) 0.4 F

162. If 0.1 M solution of glucose and 0.2 M solution of urea are placed on two sides of the Semipermeable membrane to equal heights, then it will be correct to say that -

(1) Urea will flow towards glucose solution

(2) There will be no net movement across the membrane

(3) Water will flow from glucose solution to urea.

(4) Glucose will flow towards urea solution.

163. When 7.5 g of a non-volatile non-electrolyte solute dissolved in 100 g of solvent, the boiling point of solvent increases by 0.5 K, the molar mass of solute is [K_b of the solvent = 0.52 K kg mol^{-1}]

(1) 26 g mol^{-1}

(2) 78 g mol^{-1}

(3) 52 g mol^{-1}

(4) 98 g mol^{-1}

164. Time required to decompose half of the substance for n^{th} order reaction is inversely proportional to-

(1) a^{n+1}

(2) a^{n-1}

(3) a^{n-2}

(4) a^{n}

165. In the following equilibrium

$A + 2B \rightleftharpoons 2C, \quad k_{eq.} = k_1$

$C + D \rightleftharpoons A, \quad k_{eq.} = k_2$

$4B + 2D \rightleftharpoons 2C, \quad k_{eq.} = k_3$

Find relation between k_1, k_1 and k_3.

(1) $k_1k_2 = k_3$

(2) $k_1^3 . k_2^2 = k_3$

(3) $3k_1k_2 = k_3$

(4) $k_3 = k_1^2k_2^2$

166. How many of given complexes are homoleptic complexes :-
Ferrocene, Zeises salt, Prussian blue, sodium nitroprusside, Brown ring complex.

(1) 2

(2) 3

(3) 4

(4) 5

167. The most stable complex among the following is :-

(1) $K_3[Al(C_2O_4)_3]$

(2) $[Pt(en)_2]Cl_2$

(3) $K_2[Ni(EDTA)]$

(4) $[Ag(NH_3)_2]Cl$

168. Match the list I & II and pick correct match :-

List-I Complex		List-II Structure & magnetic moment	
(A)	$[Ag(CN)_2]^-$	(i)	Square planar, 1.73 B.M.
(B)	$[Cu(CN)_4]^{-3}$	(ii)	Linear, Zero
(C)	$[Cu(CN)_6]^{-4}$	(iii)	Octahedral, Zero
(D)	$[Cu(NH_3)_4]^{+2}$	(iv)	Tetrahedral, Zero
(E)	$[Fe(CN)_6]^{-4}$	(v)	Octahedral, 1.73 B.M.

(1) A-ii, B-iii, C-v, D-i, E-iv

(2) A-i, B-v, C-iii, D-iv, E-ii

(3) A-ii, B-iv, C-v, D-i, E-iii

(4) A-i, B-iii, C-iv, D-v, E-ii

169. Which gives a pair of enantiomorphs :-

(1) $[Pt(NH_3)_4]\ [PtCl_6]$

(2) $[Co(NH_3)_4Cl_2]NO_2$

(3) $[Cr(NH_3)_5]^{+3}$

(4) $[Co(en)_2Cl_2]Cl$

170. Which of the lanthanoid ion has maximum spin magnetic moment –

(1) Pm^{+3}

(2) Pr^{+3}

(3) Nd^{+3}

(4) Ce^{+3}

171. Correct decreasing order of second ionisation enthalpy is :-

(1) F > N > O > C

(2) O > N > F > C

(3) O > F > N > C

(4) N > C > F > O

172. Correct increasing order of size is :-

(1) $Li^+ < Na^+ < F^- < Cl^-$

(2) $Na^+ < Li^+ < F^- < Cl^-$

(3) $F^- < Li^+ < Na^+ < Cl^-$

(4) $Li^+ < F^- < Na^+ < Cl^-$

173. Among following example correct decreasing EN order is :-
Cl, O, S, H, F

(1) F > Cl > O > S > H

(2) F > O > Cl > S > H

(3) F > O > Cl > H > S

(4) Cl > F > O > S > H

174. Correct increasing bond order arrangement is :-

(1) $O_2^+ < C_2^{-2} < O_2^- < He_2^+$

(2) $He_2^+ < O_2^- < O_2^+ < C_2^{-2}$

(3) $C_2^{-2} < He_2^+ < O_2^+ < O_2^-$

(4) $O_2^- < O_2^+ < C_2^{-2} < He_2^+$

175. Shape of SF_2Cl_2 molecule is :-

(1) Tetrahedral

(2) Octahedral

(3) Square pyramidal

(4) Distorted Tetrahedral

176. Which has highest dipole moment :-

(1) CH_3F

(2) CH_3Cl

(3) CH_3Br

(4) CH_3I

177. In which debye forces are present.

(1) HCl........Cl^-

(2) HCl.......Cl_2

(3) HCl........HCl

(4) Cl_2.......Cl_2

178. Maximum boiling point among group 15 hydrides will be :-

(1) SbH_3 (2) NH_3

(3) PH_3 (4) AsH_3

179. Which reagent used for Ni^{+2} ion test.

(1) Sodium hydrogen phosphate

(2) Dimethyl glyoxime

(3) Sodium nitropruside

(4) Ammonium molybdate

180. In borax bead test, bead is chemically a :-

(1) Tetraborate

(2) Orthoborate

(3) Metaborate

(4) Triborate

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